Waste Management 34 (2014) 363–369

Contents lists available at ScienceDirect

Waste Management

journal homepage: www.elsevier.com/locate/wasman

Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste

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ARTICLE INFO

Article history:
Received 12 February 2013
Accepted 28 October 2013
Available online 27 November 2013

Keywords:
Micro-aeration
Acetic acid
Butyric acid
Methane production
Protein hydrolysis

ABSTRACT

Micro-aeration is known to promote the activities of hydrolytic exo-enzymes and used as a strategy to improve the hydrolysis of particulate substrate. The effect of different micro-aeration rates, 0, 129, 258, and 387 L-air/kg TS/d (denoted as LBR-AN, LBR-6h, LBR-3h and LBR-2h, respectively) on the solubilization of food waste was evaluated at 35 °C in four leach bed reactors (LBR) coupled with methanogenic upflow anaerobic sludge blanket (UASB) reactor. Results indicate that the intensity of micro-aeration influenced the hydrolysis and methane yield. Adequate micro-aeration intensity in LBR-3h and LBR-2h significantly enhanced the carbohydrate and protein hydrolysis by 21–27% and 38–64% respectively. Due to the accelerated acidogenesis, more than 3-fold of acetic acid and butyric acid were produced in LBR-3h as compared to the anaerobic treatment LBR-AN resulting in the maximum methane yield of 0.27 L CH_4/g VS_{added} in the UASB. The performance of LBR-6h with inadequate aeration was similar to that of LBR-AN with a comparable hydrolysis degree. Nevertheless, higher aeration intensity in LBR-2h was also unfavorable for methane yield due to significant biomass generation and CO_2 respiration of up to 18.5% and 32.8% of the total soluble hydrolysate, respectively. To conclude, appropriate micro-aeration rate can promote the hydrolysis of solid organic waste and methane yield without undesirable carbon loss and an aeration intensity of 258 L-air/kg TS/d is recommended for acidogenic LBR treating food waste.

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1. Introduction

During the acidogenic phase of an anaerobic digestion, a large consortium of microorganisms that are capable of solubilizing organic substrates such as carbohydrates, proteins and lipids, work together and produce organic acids (i.e., lactic, acetic, butyric and propionic acids, etc.). Among the organic acids produced, butyric acid (HBu) and acetic acid (HAc) are known to be better precursors of methane production (Yang et al., 2003; Montero et al., 2010), especially HAc, which is regarded as the most important product, contributing 65–95% of methane production (Weber et al., 1984); in contrast lactic acid can seriously inhibit hydrolysis and methanogenesis (Zhang et al., 2007; Li et al., 2008). Therefore, conversion of organic matters into HAc and HBu in the hydrolytic-acidogenic stage will improve the overall energy yield (Yang et al., 2003) and increase the process rate. Techniques employed to enhance the solubilization of organic substrates into volatile fatty acids (VFAs) include pH control (Yu and Fang, 2003; Cysneiros et al., 2008), addition of enzymes/microbes (Cirne et al., 2008) and physicochemical/thermal pretreatments (Hartmann and Ahring, 2006).

In the recent studies, several authors documented the advantage of incorporating micro-aeration in the hydrolytic-acidogenic reactors for solubilizing the sludge and other organic substrates. Jagadabhi et al. (2010) found that the production of VFAs from the leach bed reactor (LBR) fed with energy crops increased from 2.2 to 9.0 g/L by the application of micro-aeration at 1 L/min. Zhu et al. (2009) demonstrated that the hydrolysis rate was improved by 30.9–57.7% with the provision of sufficient micro-aeration regime of 5 min aeration in every 4 h at a flow rate of 0.7 L/min as compared with those treatments with insufficient aeration (5 min/24 h) and complete anaerobic conditions. In contrast, Nguyen et al. (2007) reported that the hydrolysis rate of municipal solid waste was not improved with the aeration for 2 h at 1 L/min after 4 h interval; however, a positive effect on the performance of methanogenic phase was observed.

Although, these studies have compared the micro-aeration intensity and effectiveness in the hydrolysis of various organic substrates, the relationship between micro-aeration and acidification rate has not been completely understood in terms of the extent and types of VFAs produced in LBR. Furthermore, the composition

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http://dx.doi.org/10.1016/j.wasman.2013.10.038

of VFAs would affect the performance of subsequent methanogenic reactor. Despite the reported advantages of micro-aeration, the residual oxygen in the acidogenic leachate from LBR may inhibit the activity of methane-forming bacteria and decrease the methane yield. Therefore, the objective of the present study was to study the effect of different intermittent micro-aeration rates on the solubilization and speciation of VFAs in the anaerobic hydrolytic-acidogenic LBR reactor coupled with an up-flow anaerobic sludge blanket (UASB) reactor. Material balance based on metabolites and by-products distribution in the LBR and the methane production in the UASB of the two-stage LBR-UASB system was used as the reference for the selection of optimum micro-aeration intensity.

2. Materials and methods

2.1. Anaerobic digesters and materials

Four identical sets of laboratory scale LBR-UASB systems were used for the experiments. The working volume of acidogenic LBR and methanogenic UASB were 4.6 L and 10.0 L respectively. The detailed configurations of reactors were presented in our previous report (Xu et al., 2011). Synthetic food waste (FW) with a total solids (TS) content of 38.5 ± 1.0% with volatile solids (VS) content of 97.1 ± 0.2% (VS/TS) was used as the substrate (Wong et al., 2009). Anaerobically digested sludge (ADS) collected from the Shek Wu Hui wastewater treatment plant, Hong Kong, with 5.2% TS and 86.1% VS/TS was used as the inoculum for acidogenic reactor.

2.2. Experimental design

Experiments were performed for a total of 17 days at 35 °C with two replicates for each treatment to obtain reproducible results. Intermittent micro-aeration was applied using an air pump connected to the bottom of the LBRs with a flow rate of 1.0 L/min; and the operation was controlled using timers. Four micro-aeration frequencies were evaluated, i.e., aeration for 12 min every 6 h (LBR-6h), 3 h (LBR-3h), 2 h (LBR-2h) and no aeration (LBR-AN) resulting in corresponding micro-aeration rates of 129, 258, 387 and 0 L-air/kg TS/d, respectively.

Each LBR was initially packed with a mixture of 1.0 kg FW and 0.2 kg of anaerobically digested sludge (ADS). Additionally, 10% wood chips (in volume basis) were added as the bulking agent to facilitate the liquid leaching (Xu et al., 2011). Then 1.0 L of tap water was added into each LBR on Day 0. Leaching occurred naturally and the leachate was collected at the bottom of the reactors daily. Half of the collected leachate was diluted with equal volume of tap water and adjusted to pH 6.0 using sodium carbonate before recycling back into the LBR. The remaining half of the collected leachate was fed into the UASB reactor for methanogenesis.

2.3. Analytical methods

The volume of biogas produced from the UASB reactor was measured using a wet gas flow meter (BSD-0.5, Shanghai, China) and concentration of methane in the biogas generated was determined using a gas chromatograph (GC-HP7890) equipped with a thermal conductivity detector. Total solids (TS) and volatile solids (VS) content of the FW and seed sludge were determined by oven drying at 105 °C for 24 h and igniting at 550 °C for 16 h in a muffle furnace, respectively. Total organic carbon (TOC) content of the FW and digestate were determined by dichromate oxidation method (Nelson and Sommer, 1982). The total Kjeldhal nitrogen (TKN) content of the FW and digestate were determined by digestion and spectrophotometric determination of ammonium-nitrogen (NH_4^+-N) according to the APHA Standard Method 4500-N_{org} C (APHA, 2005).

Acidogenic leachate from the LBR and methanogenic leachate from the UASB reactor were collected everyday and analyzed. The pH and ORP were measured with a pH/ORP meter (Orion 920, Thermo). The concentration of chemical oxygen demand (COD), NH_4^+-N and TKN were analyzed according to the Standard Methods 5220C, 4500-N_{org} C and 4500-NH_3 F, respectively (APHA, 2005). For the analysis of VFAs, leachates were filtered through 0.45 μm cellulose acetate membrane filter paper and 0.9 mL was transferred to GC vials with the addition of 0.1 mL formic acid. The concentrations of acetic acid, propionic acid, *n*-butyric acid, *iso*-butyric acid, *n*-valeric acid and *iso*-valeric acid were determined using a HP 6890 Series gas chromatograph (Hewlett Packard) equipped with flame ionization detector and Econo-Cap EC1000 (15 m × 0.53 mm × 1.20 μm) column. The sample injection volume was 1 μL. The temperatures of the injector and detector were maintained at 200 °C and 250 °C, respectively, while the column temperature was programmed with an initial temperature of 120 °C for 5 min, increasing to 160 °C at a rate of 5 °C/min, and holding at 160 °C for 5 min to ensure complete VFAs volatilization. The sum of acetic acid, propionic acid, *n*-butyric acid, *iso*-butyric acid, *n*-valeric acid and *iso*-valeric acid concentrations are reported as total VFAs (tVFA).

3. Results and discussion

3.1. Performance of LBRs with varied micro-aeration intensities

3.1.1. Changes in pH, ORP and volume of leachate

Recycling of acidogenic leachate back to LBR and adjusting the pH of the recycling leachate to 6.0 enhanced the hydrolysis as shown in our previous study (Xu et al., 2011). Therefore, 50% of the acidogenic leachate was mixed with equal volume of tap water, adjusted the pH using sodium carbonate and then recycled back to the LBR in order to recycle part of the hydrolytic enzymes and reduce the acid crisis. The volume of leachate produced in LBR ranged from 480 to 520 mL for most of the period (Fig. 1a), and the differences in decomposition rate among the four treatments influenced the percolating property resulting in varied levels of leachate production.

The pH of LBRs ranged from 3.8 to 4.2 in the first 3 days, and then gradually increased. During the whole digestion process, pH of LBRs' leachate that was recycled to LBR was daily adjusted to 6.0, thus the pH fluctuated around 6.0. The pH in four LBRs was approaching the range of 5.5–6.5 after about one week of digestion. At the beginning of digestion, ORP of digestate was around 56 mV and decreased gradually along the digestion period. By the end of digestion, ORP decreased to the level between −200 mV and 300 mV. The ORP values showed a contrasting profile to that of the pH profile as reported by Zhu et al. (2009). ORP is negatively associated with pH if only the half reaction $[H^+]/[H_2]$ is accounted. In LBRs with micro-aeration, ORP increased to 125–215 mV on the second day and then decreased immediately to the level between −150 mV and −300 mV in the following one week, whereas ORP in LBR-AN decreased gradually right at the beginning of digestion. The initial increase followed by a rapid decrease of ORP can be attributed to the stimulation of facultative microorganisms, which leads to the accelerated consumption of oxidants. Nevertheless, the ORP level in LBR-2h was higher than others indicating comparatively higher level of oxidants in this reactor.

3.1.2. Changes in COD, TKN and NH_4^+-N contents of leachate

As presented in Fig. 2a, COD concentrations of the first day digestion ranged from 7.9 to 9.3 g/L, which was followed by the

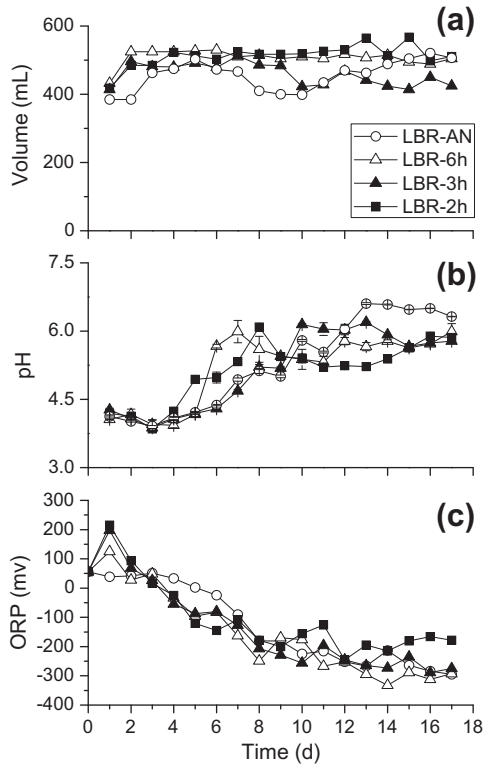


Fig. 1. Characteristics of discharged leachates from four LBRs: (a) volume, (b) pH and (c) ORP.

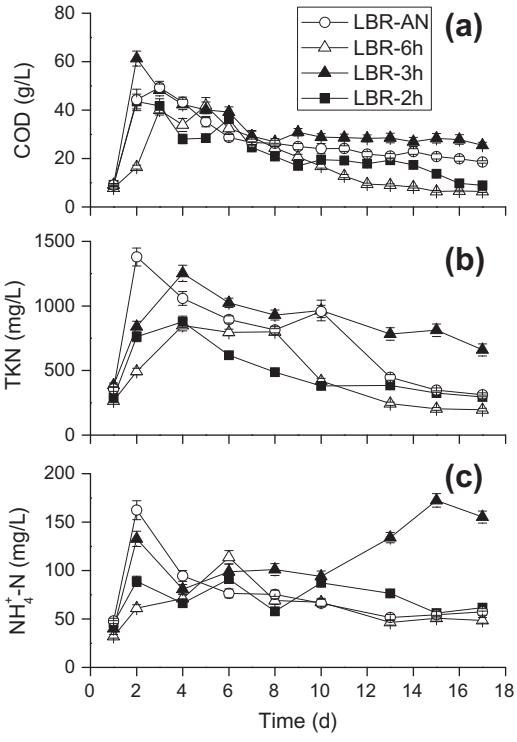


Fig. 2. Characteristics of discharged leachates from four LBRs: (a) COD concentration, (b) TKN concentration and (c) NH_4^+-N concentration.

peak values observed on the second or third day, i.e. COD concentrations of 41.8, 61.3, 43.5 and 49.4 g/L, respectively, for LBR-AN, 6 h, 3 h and 2 h. The relatively low concentration of soluble hydrolysate in leachate indicates that the stagnation period of hydrolysis was about one day. A similar pattern was found for TKN and NH_4^+-N leaching also. However, the trend varied among four treatments. COD concentration in LBR-3h exhibited the peak of 61.3 g/L on the second day and the highest levels ranged 25.4–30.8 g/L during the subsequent digestion period. Concentration of COD in LBR-AN reached the maximum value of 49.4 g/L on the third day and maintained the second highest level during the latter phase; while LBR-2h and LBR-6h ranked the third and fourth. Similar trend was found for TKN also as presented in Fig. 2b. TKN concentrations of LBR-3h were in the range of 660–1252 mg/L, while LBR-2h and LBR-6hranged between 200 and 878 mg/L.

During the initial stage of digestion, difference in NH_4^+-N concentrations among the four treatments was insignificant, but there was a distinct increase in LBR-3h in the late period of digestion indicating that the aeration intensity may exert effects on the deamination or related nitrogen transformation. The tradeoff between the higher protein decomposition rate of 70.8% (Table 2) and the lower concentrations of TKN and NH_4^+-N as detected in leachate from LBR-2h, indicates that some soluble NH_4^+-N might have escaped from the reactor as ammonia with air flushing, or part of the nitrogen might have transformed into nitrate/nitrite through aerobic nitrification. Since the hypothesis of ammonia emission through air flushing is not supported by the fact that the pH level in LBRs was lower than 6.5, which is not favorable for ammonia volatilization, nitrification could be the major factor, but additional studies are required to confirm this reasoning.

3.2. VFAs production and speciation in acidogenic LBRs

3.2.1. VFAs production

As shown in Fig. 3, at the beginning of digestion, no significant differences in the tVFA concentrations were found among four LBRs, which indicate that the performance of LBR at the early stage was mainly influenced by the inoculum-born microbial activity. On the second day of digestion, nearly 3-fold higher concentration of tVFA was observed in LBRs with micro-aeration than that of control (LBR-AN). Afterwards, concentrations of tVFA in LBR-AN was the lowest throughout the digestion process. As shown in Fig. 3f, the specific yields of VFAs in LBR-AN was 0.09 g COD/g VS_{added}, which was lower than LBR-6h, 3h and 2h (0.26, 0.40 and 0.18 g COD/g VS_{added}, respectively). A positive effect of micro-aeration on the VFAs production was also reported by Jagadabhi et al. (2010) using grass-silage as substrate. Limited oxygen concentration would result in the accumulation of reduced organic matter due to the poor tricarboxylic acid cycle (Jensen and Michelsen, 1992). Tiquia et al. (1996) also found that regulated limitation of oxygen could result in the accumulation of fermentative metabolic products under facultative environment. Thus the application of micro-aeration could enhance the acid fermentation and promote the production of VFAs compared with anaerobic treatment. Furthermore, the optimum ORP range for acid fermentation was known to be −100∼−250 mV. As seen in Fig. 1c, ORP level in LBR-AN was higher than −100 mV before Day 7, whereas ORP in other three LBRs decreased to lower than −100 mV from Day 4, which may lead to the lower level of tVFA in LBR-AN.

Nevertheless, tVFA levels in LBR-6h, 3h and 2h were influenced by different aeration intensities. The highest tVFA level of 17.5–21.8 g COD/L was observed in LBR-3h during the period of Day 6 to 17. In case of LBR-6h, the concentrations of tVFA peaked on Day 4 (18.3 g COD/L) but decreased gradually to 6.23 g COD/L by the end of operation. Furthermore, tVFA level in LBR-2h was much lower than LBR-6h and LBR-3h, indicating the negative effects of higher levels of aeration. Johansen and Bakke (2006) also observed that the accumulation of VFAs in the digester with micro-aeration of 15 s per min was only half of the anaerobic reactor. These

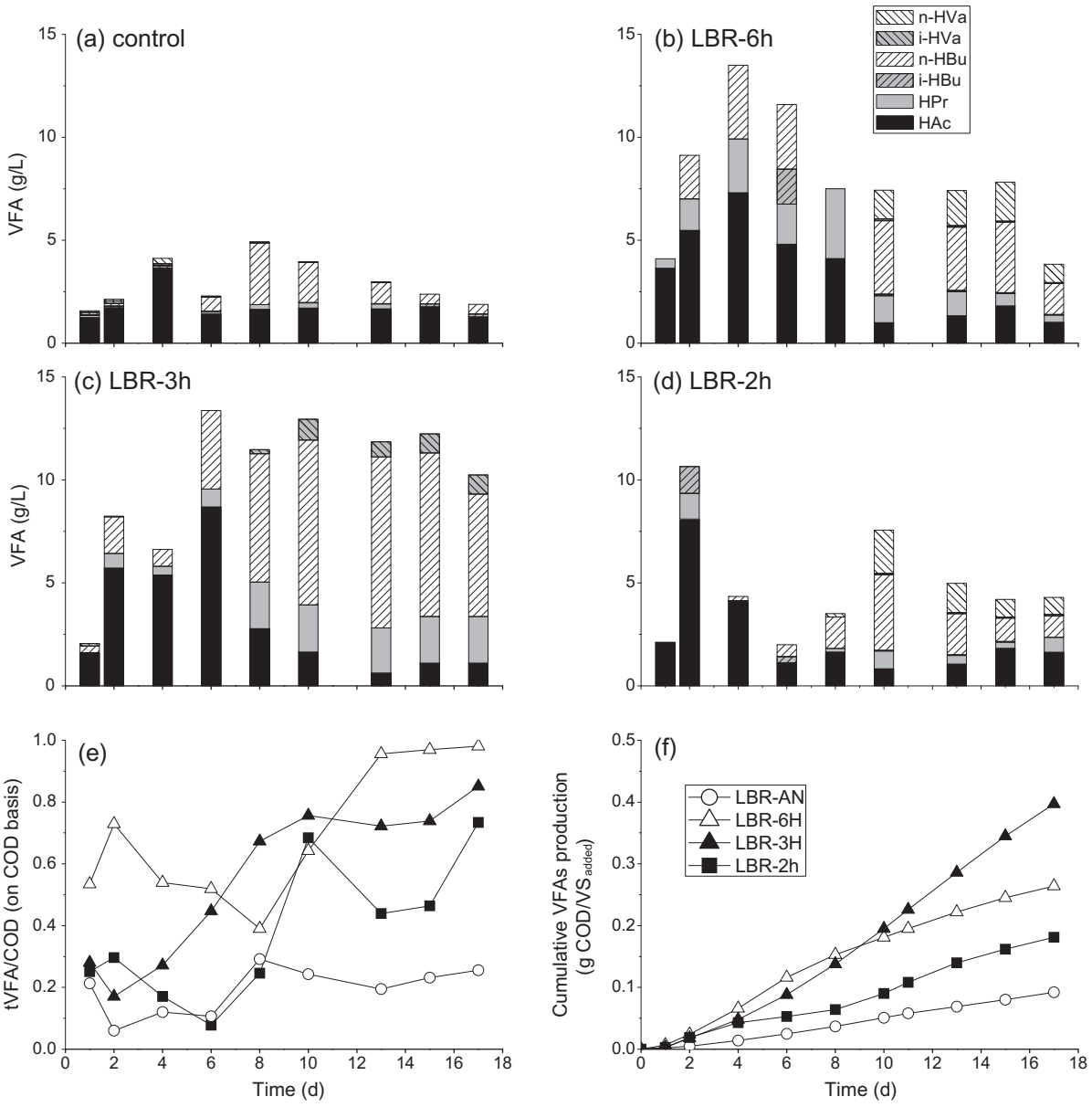


Fig. 3. VFAs levels of leachates from (a) LBR-AN, (b) LBR-6h, (c) LBR-3h, and (d) LBR-2h, (e) the ratio of tVFA to COD and (f) cumulative production of VFAs.

negative results might be due to the increased generation of aerobic bacteria and aerobic respiration leading to the lower level of intermediate products. Thus more VFAs were generated in LBR-3h with a comparatively low aeration rate, while more VFAs were consumed in LBR-2h with higher aeration rate.

3.2.2. VFAs speciation

HAc and HBu were the two main species of VFAs produced in all LBRs as seen in Fig. 3a and b. This result is consistent with the previous reports of Han and Shin (2004) and Komemoto et al. (2009) using FW as the substrate. In present study, the pH values of four LBRs were in the range of 4.5–6.5, which would be favorable for the acidogenesis. Solubilization of carbohydrates increase substantially and results in an increase in the generation of all acids when pH increases to >4.5; whereas, when the pH increases to >6.5, HBu production essentially ceases (Fang and Liu, 2002). Since the pH of LBRs was below 6.5 in this experiment, HBu was produced all through the experiment in consistent with the earlier reports. During the initial 5 days of digestion, HAc predominated in all LBRs, which accounted for 66.8–95.0% of tVFA in the leachates. Meanwhile, the ratios of tVFA/COD in the first day ranged from 21% to 53%, which were relatively low compared to the values of 73–98% at the end of digestion (Fig. 3e). This is probably due to the contribution of formic acid and lactic acid, which were reported to account for ~50% of the total soluble products (Zhu et al., 2009). Therefore, although the formic acid and lactic acid were not analyzed in this study, it can be reasonably assumed that these species may contribute to the remaining COD.

After Day 5, the production of HAc was reduced while the productions of HBu and propionic acid (HPr) were increased correspondingly. HAc constituted about 31.3–43.8% of tVFA produced in the Control during the period of Day 8–17; whereas, only 9.8–17.5%, 3.3–16.4% and 6.7–30.7% of HAc were observed in LBRs with aeration intensities of 12 min/6 h, 3 h and 2 h, respectively. As summarized in Table 1, the fraction of HPr increased from 3% of tVFA in LBR-AN to 12%, 7% and 5% in LBR-6h, 3h and 2h, respectively; the production of HVa also increased from 2% of tVFA in LBR-AN to 3–11% in LBRs with micro-aeration. Conclusively,

Table 1
VFAs production and speciation in LBRs with different micro-aeration intensities.

		LBR-AN	LBR-6h	LBR-3h	LBR-2h
Specific VFAs yield in LBR (g COD/g VS_{added})		0.09	0.26	0.39	0.19
VFAs compositionin LBR leachate (% of tVFA, on COD basis)	HAc	65	61	58	62
	HPr	3	12	7	5
	HBu	30	22	32	22
	HVa	2	5	3	11

Table 2
Effect of micro-aeration on the hydrolysis efficiency and methane yield of food waste.

	LBR-AN	LBR-6h	LBR-3h	LBR-2h
Micro-aeration intensity (L-air/kg TS/d)	0	129	258	387
Total volume of leachate (L)	8.3	9.8	9.9	10.0
Degree of hydrolysis (%)[a]	62.3	60.1	66.3	78.1
Decomposition of carbohydrate (%)	65.4	69.1	79.6	83.0
Decomposition of protein (%)	43.2	42.7	59.7	70.8
Specific COD yield in LBR (g COD/g VS)	0.87	0.67	1.05	0.66
Methane yield in UASB (L CH_4/g VS_{added})	0.23	0.17	0.27	0.19

[a] Degree of hydrolysis = $1 - (VS_{residual}/VS_{in})$.

micro-aeration accelerated the fermentation as well as shifted the metabolism from HAc production to HBu and HPr production. The decomposition of easily degradable carbohydrates is reported to be associated with the production of HAc, while the hardly degradable carbohydrates increase the production of HPr (Lü et al., 2009).

HAc and HBu are the desirable VFAs in the acidogenic phase of anaerobic digestion to maximize methane production (Yang et al., 2003). In contrast, accumulations of HPr and HVa would be limiting factors for methane generation. In the present study, although the proportions of HPr and HVa were increased in reactors with aeration, the production of HAc and HBu were also increased due to the higher yield of tVFA. Micro-aeration intensity of 258 L-air/kg TS/d in LBR-3h increased the production of VFAs; furthermore, the summed content of HAc and HBu was at a comparable level (90%) as compared to LBR-AN (95%). Nevertheless, the absolute content of HAc increased significantly from 0.06 of LBR-AN to 0.23 g COD/g VS_{added} of LBR-3h, which could be favorable for methane generation.

3.3. Overall performance of LBR-UASB System

The hydrolysis rate and methane yield of FW under different micro-aeration intensities are presented in Table 2. The degree of hydrolysis for each treatment is calculated based on the VS removal, while the decomposition efficiencies of carbohydrate and protein are calculated according to the TOC and TKN removal. Degrees of hydrolysis in LBR-3h and LBR-2h were 66.3% and 78.1%respectively, showing improvement when compared with the control LBR-AN (61.2%). Furthermore, carbohydrate and protein hydrolysis efficiency were enhanced from 65.4% and 43.2% in LBR-AN to 70.8% and 78.1% in LBR-2h. The positive impacts of micro-aeration on the hydrolysis efficiency of carbohydrates and protein have been reported by several researchers (Hasegawa et al., 2000; Jagadabhi et al., 2010; Lim and Wang, 2013). Hasegawa et al. (2000) also reported that the hydrolysis rate and methane yield from sewage sludge was improved by ~50% after micro-aeration treatment. The improved hydrolysis could be attributed to the stimulation of fermentative bacteria to excrete hydrolytic enzymes (Nguyen et al., 2007; Zhu et al., 2009). In the anaerobic fermentation of organic waste, the facultative bacteria can consume oxygen and induce anaerobic conditions, while the strict anaerobes supply metabolites for facultative bacteria (Ueno et al., 2006; Li and Fang, 2007). Thus, regulating the intermittent aeration or micro-aeration could stimulate the effective co-existence of facultative bacteria and anaerobes (Sang et al., 2009). However, the protein decomposition (42.7%) and degree of hydrolysis (60.1%) in LBR-6h were comparable with the control LBR-AN; probably related to the unstable enzyme activity under the insufficient aeration (Zhu et al., 2009). Furthermore, the soluble hydrolysate contents as indicated by the specific COD yield did not correlate positively in all the treatments. LBR-3h with an intermediate aeration rate produced the highest specific COD yield of 0.78 g COD/g VS_{added}, while the insufficient aeration in LBR-6h and excess aeration in LBR-2h resulted in lower COD yields of 0.45 and 0.52 g COD/g VS_{added}, as compared to the control (0.58 gCOD/g VS_{added}). With a decrease in intermediate soluble products, the final methane yields in the UASB decreased correspondingly. As shown in Table 2, the highest methane yield of 0.27 L CH_4/g VS_{added} was achieved with LBR-3h and the lowest yield of 0.17 L CH_4/g VS_{added} was observed for LBR-6h; and this trend reflects the specific COD production in the LBR.

The optimum micro-aeration intensity of 258 L-air/kg TS/d in the present experiment was comparable to the study of Zhu et al. (2009) with sufficient aeration intensities of 147 and 442 L-air/kg TS/d. However, these values were much higher than that reported by Johansen and Bakke (2006), and Lim and Wang (2013), which were about 33 L air/kg TS/d and 57 L air/kg TS/d. This difference might be related to the way of intermittent aeration applied in Zhu et al. (2009) and in the present study, while continuous micro-aeration with a low flow rate was applied in other two references. In our study, 12 L of air passed through LBR during each time of ventilation, nevertheless it is difficult to completely consume this volume of oxygen within 12 min. Therefore the practical oxygen supply should be much less. The micro-aeration intensity of our study was estimated to be 16, 32 and 48 L air/kg TS/d according to the working volume of reactor. Then the micro-aeration intensities among above studies are comparable. Furthermore, oxygen supply in micro-aeration treatments ranged from 12% to 35% of feed COD, which was comparable to 16% reported by Botheju et al. (2010).

3.4. Carbon balance based on metabolites and by-products distribution

Hydrolysis breaks down the particulate substrate into soluble substrates such as sugars; and the extracellular enzymes play a crucial role in the hydrolysis. Subsequently, the hydrolyzed particulate substrates (S_{Hy}) are utilized by the acidogens, to produce VFAs and gases (CO_2 and H_2). The total amount of $S_{Hy,TOC}$ can be expressed based on the carbon balance according to the following equation,

$$S_{Hy,TOC} = X_{Bm,TOC} + S_{In,TOC} + \text{Biogas}_{TOC} \quad (1)$$

where $S_{In,TOC}$ is the soluble intermediate products in the effluent from acidogenic reactor on the basis of TOC, $X_{Bm,TOC}$ is the biomass proliferation using hydrolyzed substrate. $S_{Hy,TOC}$ is calculated from the discrepancy of TOC content between the initial and final digestate.

The hydrolyzed nitrogenous organic matters ($S_{Hy,N}$) is distributed in the soluble intermediate products containing nitrogen ($S_{In,N}$) and the biomass generation ($X_{Bm,N}$) as expressed in the following equation,

$$S_{Hy,N} = X_{Bm,N} + S_{In,N} = (N/C)_{XBm} \cdot X_{Bm} + (N/C)_{SIn} \cdot S_{In} \quad (2)$$

where $(N/C)_{XBm}$ and $(N/C)_{SIn}$ are the nitrogen to carbon ratio of biomass and intermediate products, respectively. $S_{Hy,N}$ is calculated from the difference of TKN content in digestate before and after digestion, while $S_{In,N}$ is the cumulative TKN content eluted in the leachate. Thus $X_{Bm,N}$ can be evaluated according to Eq. (2), and a typical biomass composition ($CH_{1.8}O_{0.5}N_{0.2}$) was referred to calculate the content of X_{Bm} as suggested by Johansen and Bakke (2006). The distributions of residual substrate (S_{Re}), hydrolyzed substrate (S_{Hy}), soluble intermediate products (S_{In}), biogas and biomass (X_{Bm}) in LBRs are shown in Fig. 4a. In the succeeding methanogenic stage of digestion, acetotrophic methanogens and hydrogenotrophic methanogens make use of VFAs, CO_2 and H_2 to produce methane. However, a part of organic matters are also used for biomass generation or flow out with effluent. The ratio of methane production to the influent $S_{In,N}$ in each treatment is presented in Fig. 4b.

As illustrated in Fig. 4a, S_{Re} decreased with the increasing micro-aeration rate, and S_{Hy} increased correspondingly from 65.5% in LBR-AN to 83.0% in LBR-2h. However, the S_{In} did not follow the same trend as that of the total hydrolysis rate. With an intermediate aeration rate, the S_{In} of LBR-3h accounted for 75.7% of the total amount of S_{Hy} and it gave the maximum COD yield correspondingly. When aeration rate increased to 12 min/2 h, the S_{In} only accounted for 48.7% of S_{Hy}. Similarly, the carbon loss due to respiration under excess micro-aeration has also been reported by Charles et al. (2009) and Jenicek et al. (2008) with the substrate of organic solid wastes. Moreover, the increased biomass generation was found in LBRs with the higher aeration rate. As shown in Fig. 4a, LBR-2h showed the highest yield of bacterial growth of 18.5% ($X_{Bm}/COD_{hydrolyzed}$), while LBR-AN, LBR-6h and LBR-3h had much lower levels of 0.9, 6.6 and 3.3%, respectively. Akutsu et al. (2009) found that the yield of bacterial growth tended to decrease with an increase in the total concentration of undissociated organic acids and attributed this phenomenon to the decreased availability of ATP. This explained why the biomass generation in LBR-3hwas much lower than that of LBR-2h. Therefore, the tradeoff between the high hydrolytic efficiency and the lower level of organics being available for methanogenesis should not be neglected, because the

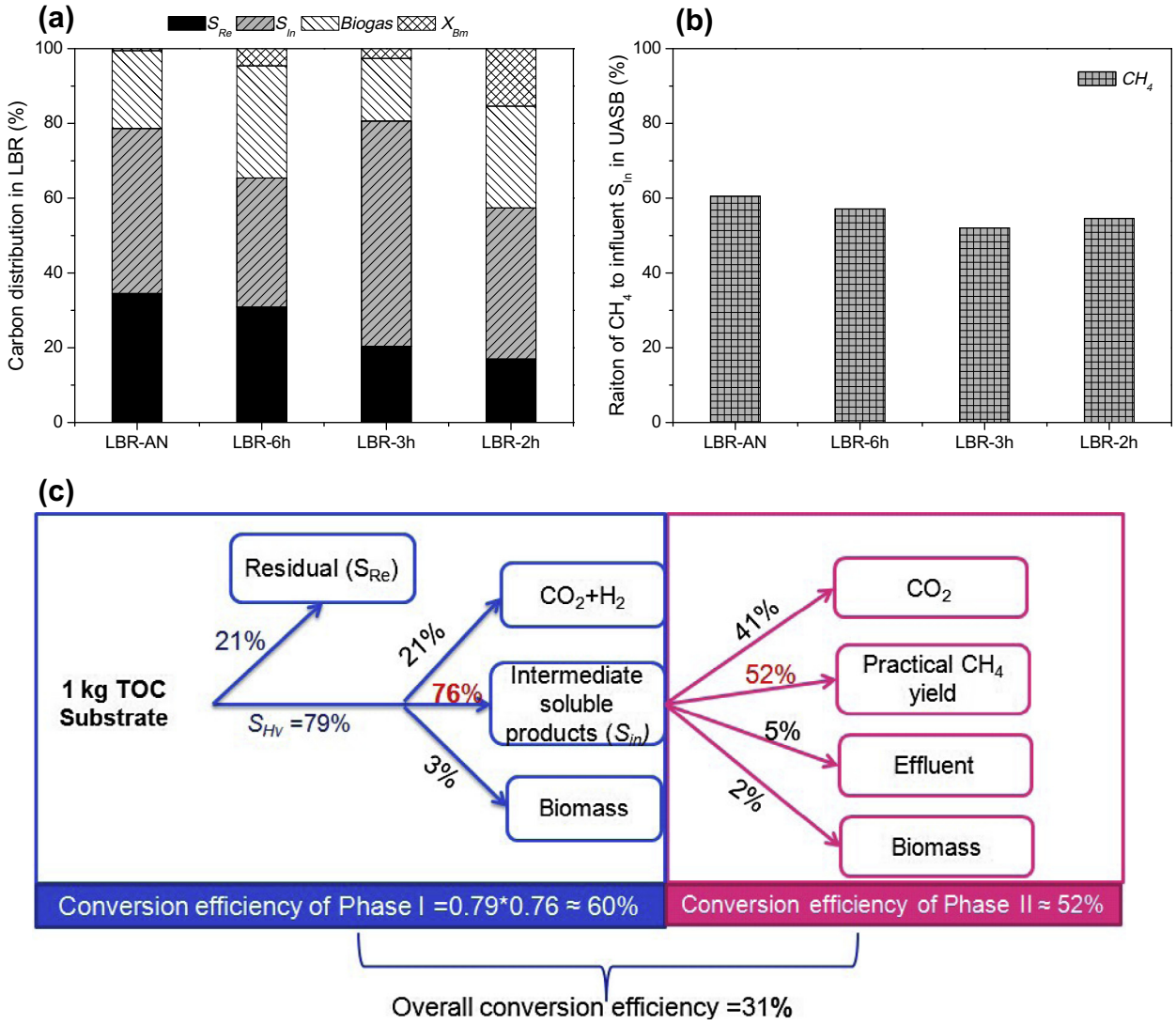


Fig. 4. (a) Carbon distribution in LBRs with various micro-aeration intensities, four parts are presented: residual substrate (S_{Re}), hydrolyzed substrate (S_{Hy}), soluble intermediate products (S_{In}), biogas and biomass (X_{Bm}); (b) the efficiency of converting influent S_{In} into CH_4 in UASB; (c) carbon flow in LBR-3h connected with UASB system.

amount of carbon loss though CO_2 or microbial biomass is significant.

In the methanogenic reactor, the efficiency of transforming S_{In} into CH_4 was quite similar in all four treatments, in the range of 54.6–60.5% (Fig. 4b). Thus, the methane yield in UASB was positively correlated to the amount of S_{In} obtained in acidogenic LBR. The highest methane yield of 0.27 L CH_4/g VS_{added} was achieved in UASB connected to LBR-3h with the highest COD concentration in leachate.

4. Conclusions

This study evaluated the effects of four micro-aeration intensities on the hydrolysis and methane yield of food waste in a two-phase anaerobic digester. The optimum micro-aeration intensity was 258 L-air/kg TS/d in LBR-3h that accelerated COD leaching and enhanced methane production by 5% and 18%, respectively. The generation of HAc and HBu in LBR-3h was more than three times higher than the treatment with anaerobic treatment LBR-AN resulting in the maximum methane yield of 0.27 L CH_4/g VS_{added}. The performance of LBR-6h with inadequate aeration was similar to LBR-AN with a comparable hydrolysis rate and decreased methane yield. Nevertheless, higher aeration intensity in LBR-2h was also unfavorable for methane yield due to the oxidation of intermediate soluble products, which decreased the specific COD production to 0.19 L CH_4/g VS_{added} although with a high degree of hydrolysis. Results indicate that optimal rate of micro-aeration can promote the solubilization of solid organic waste without undesirable carbon loss and the aeration of 258 L-air/kg TS/d is advantageous for food waste.

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